Q1. You are building a hospital system. Identify at least 3 classes you would create and explain what instance variables and methods each would have.

Patient Class:
A Patient class could contain instance variables such as patientId, name, age, and disease. Methods could include registerPatient(), updateDetails(), and displayPatientInfo(). This class would store and manage patient-related information.

Doctor Class:
A Doctor class might have variables like doctorId, name, specialization, and experience. Methods such as diagnosePatient(), prescribeMedicine(), and displayDoctorInfo() could be included. It would represent doctors working in the hospital.

Appointment Class:
An Appointment class could contain appointmentId, patientName, doctorName, and appointmentDate. Methods like bookAppointment(), cancelAppointment(), and showAppointmentDetails() would manage appointment records.

Q2. Why is it better to use private fields with getters/setters instead of making all fields public? Give a real-world scenario where this matters.

Private fields provide data hiding and protect object data from unauthorized modification. Getters and setters allow us to validate data before changing it. For example, in a bank account system, the balance field should not be public because anyone could directly set it to a negative value. Using a setter allows checking whether the amount is valid before updating the balance.

Q3. When would you choose a parameterized constructor over a default constructor? When would you use both together?

A parameterized constructor is used when objects need to be initialized with specific values at the time of creation. A default constructor is useful when no initial values are provided and default values are acceptable. Both can be used together to provide flexibility. For example, a Student object may either start with default values or be created with a specific roll number, name, and marks.

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Q5. A static variable is shared across all objects. Give one scenario where this is useful, and one scenario where it could cause a bug.

A static variable is useful when information should be common to all objects. For example, a studentCount variable can keep track of the total number of Student objects created. However, using a static variable incorrectly can cause bugs. For instance, if a bank account balance were made static, changing the balance of one account would affect all accounts, which would produce incorrect results.